Supplementary Material

Reliability of Short Molecular Dynamics Runs as Screening Labels: Noise Ceilings, Time Dependence and Adaptive Replicate Allocation

Francisco Hernando-Gallego, Dimitar Vasilev-Vasilev, Diego García-Santos, Alfonso Campo-Camacho, Alberto Gómez-Gómez

S1 Data set

Table S1: Data used in this study. All systems: LiTFSI in an amorphous polymer, PCFF+, 353 K.

	5 ns runs	50 ns runs
Trajectories	6,137	133
MSD curves used (Li^+, TFSI)	100 lags each	100 lags each
Longest lag in the curve	5 ns	50 ns
Systems also simulated at the other length	133	133
Systems simulated twice at this length	94	n/a

Raw trajectories of the 133 long runs were read as a stream, keeping only ion positions (lithium and the nitrogen of TFSI), the polymer atoms of the types that the platform lists as solvating, and the system centre of mass. That list of types is the same for every system while the type numbering is not, so the kept polymer atoms are not always those that coordinate lithium (often they are hydrogen or carbon atoms); they are used only to follow the mobility of the polymer. The origin-averaged MSD computed from them reproduces the curves distributed by the platform to within a relative deviation of 5×10^{-7} in every run, for lags up to 80% of the run. Without subtracting the centre-of-mass motion, the MSD at a lag of 25 ns would be inflated by a median factor of 1.2 and by up to a factor of 34 (more than 10% in 63% of the runs); the largest factors occur in the slowest systems, where a drift of the whole system by a few nanometres is comparable to the displacement of the lithium ions.

S2 Sensitivity of the regime census and of the lag dependence

Table S2: Sensitivity of the regime census to the lag window used to fit the effective exponent. The main text uses 0.2 to 0.6.

	5 ns runs			50 ns runs		
Fit window (fraction of run)	Median β	$\beta < 0.7$	$\beta \geq 0.9$	Median β	$\beta < 0.7$	$\beta \geq 0.9$
0.1 to 0.5	0.38	99.7%	0.0%	0.60	81.2%	0.0%
0.2 to 0.6	0.43	97.8%	0.03%	0.65	69.2%	3.8%
0.3 to 0.7	0.48	94.9%	0.07%	0.69	54.9%	5.3%
0.4 to 0.8	0.51	87.4%	0.75%	0.73	45.9%	14.3%

Table S3: Sensitivity of the lag dependence in the 133 long runs to the reference lag τ_{ref}. The main text uses 25 ns.

	Median $D_{\mathrm{app}}(\tau)/D_{\mathrm{app}}(\tau_{\mathrm{ref}})$			Spearman ρ		
τ_{ref} (ns)	$\tau = 1$	$\tau = 5$	$\tau = 10$	$\tau = 1$	$\tau = 5$	$\tau = 10$
15	4.43	1.65	1.18	0.95	0.98	0.99
20	4.93	1.83	1.30	0.93	0.97	0.99
25	5.23	1.95	1.40	0.92	0.96	0.98
30	5.56	2.05	1.47	0.92	0.96	0.98
35	5.76	2.14	1.53	0.92	0.96	0.97

Table S4: Apparent diffusivity and ranking as a function of the observed lag in the long runs, relative to a lag of 25 ns (curve-based).

Lag (ns)	Median ratio	10th to 90th percentile	Within 25%	Spearman ρ
0.5	8.86	4.70 to 12.47	0.0%	0.91
1	5.23	3.12 to 7.26	0.0%	0.92
2	3.34	2.15 to 4.32	0.0%	0.94
5	1.95	1.47 to 2.40	0.8%	0.96
10	1.40	1.18 to 1.63	16.5%	0.98
15	1.19	1.08 to 1.31	73.7%	0.99
20	1.07	1.03 to 1.12	100.0%	1.00

S3 Slowdown within the long runs

Each long run is split into ten consecutive 5 ns segments, and the first segment is compared with the five late ones (25 to 50 ns). Mobilities are end-point estimates over each segment, after subtracting the centre-of-mass motion, for lithium, for the nitrogen of TFSI and for the polymer atoms kept from the raw trajectory (125 systems that have any). Production is at constant volume (NVT); the potential energy and pressure are the segment means of the 100 ps averages written by the simulations. Ion clustering uses the lithium-O(TFSI) clusters computed in every frame during the simulations: the size of the cluster that contains each lithium, and the fraction of lithium ions whose cluster contains no anion oxygen; the number of anion nitrogens around each lithium uses a cutoff at the first minimum of the lithium-nitrogen radial distribution function of each system.

Table S5: What changes between the first 5 ns segment of each long run and the late segments (133 runs). Mobilities are ratios first/late (median and bootstrap 95% interval); the other quantities are differences first − late. "Late segments" gives the direction of the change, "Systems" the share of runs in which it occurs, and ρ the Spearman correlation, across systems, with the slowdown of lithium.

Quantity	First vs late (95% CI)	Late segments	Systems	Screening	Literature	ρ with Li slowdown (95% CI)
Lithium diffusivity (ratio)	1.63 (1.52 to 1.81)	slower	93%	1.80	1.28	n/a
Anion diffusivity (ratio)	1.46 (1.39 to 1.61)	slower	89%	1.57	1.32	0.70 (0.60 to 0.77)
Polymer-atom mobility (ratio)	1.43 (1.37 to 1.51)	slower	96%	1.50	1.26	0.68 (0.57 to 0.76)
Potential energy (kcal mol^{-1} per atom)	0.014 (0.013 to 0.015)	lower	100%	0.015	0.011	0.58 (0.46 to 0.68)
Pressure (atm)	69 (50 to 79)	lower	83%	79	34	0.30 (0.13 to 0.45)
Size of the cluster of a Li (ions)	0.37 (0.32 to 0.44)	smaller	85%	0.37	0.37	−0.03 (−0.20 to 0.15)
Fraction of Li without anion	−0.03 (−0.04 to −0.02)	larger	75%	−0.02	−0.05	0.15 (−0.02 to 0.31)
Anion N around each Li	0.06 (0.04 to 0.08)	fewer	82%	0.07	0.04	−0.06 (−0.24 to 0.10)

S4 Run-to-run reproducibility

Table S6: Reproducibility of a single 5 ns run, from 94 systems simulated twice from different initial structures. Intervals by bootstrap over systems.

Estimator	σ_{run} (log)	Typical factor per run	Rank corr. between runs	Reliability r (95% CI)	ICC(1) (95% CI)
End-point	0.35	1.43	0.48	0.47 (0.31 to 0.62)	0.53 (0.33 to 0.66)
Origin-averaged, lag $T/2$	0.28	1.32	0.56	0.58 (0.44 to 0.70)	0.62 (0.45 to 0.74)

Two runs of the same system differ by more than a factor of 1.5 in 38% of the replicated systems. The replicated subset is not range-restricted (spread of $\ln D_{\mathrm{app}}$ 0.44 against 0.49 in the whole screening set). The effective exponent of an individual 5 ns run is not reproducible (noise 0.15, rank correlation between runs -0.07): it characterises the population, not a single system.

Structural and dynamic noise. The dynamic noise is estimated from the 96 screening pairs (the 5 ns run against the first 5 ns of the long run from the same structure) and the total from the 94 replicated systems. The short run of a pair is the original replica of the same system, so the two differences are correlated (correlation 0.53 on the 92 systems that have both). Resampling those systems jointly, the structure accounts for 57% of the run-to-run variance of the end-point label (44 to 69%; separate resampling of the two sets, as in the main text: 55%, 31 to 72%) and 52% (33 to 66%) of that of the origin-averaged label.

Batches. In the data set the original screening runs, the replicas and the long runs belong to three different batches. A batch effect would appear as a shift between them or as a different spread across systems. Neither is seen for the end-point label: the mean ratio original to replica is 1.05 (95% interval 0.95 to 1.17, $p = 0.31$), the mean ratio of the short run to the first 5 ns of its long run is 1.00 (0.93 to 1.06, $p = 0.89$), and the spread of $\ln D_{\mathrm{app}}$ across systems is 0.50 for the originals and 0.53 for their replicas (Levene $p = 0.67$; 0.47 in the rest of the screening set). The noise estimates use the standard deviation of the differences, which excludes a constant offset; without removing it the reliability is 0.47, unchanged. Batch and initial structure cannot be fully separated, because every replica is both new and from a later batch.

S5 Lithium solvation and ion aggregation

For each of the 133 long runs (96 screening and 37 literature polymers), the polymer atoms kept from the trajectories (Section S1) within 3.4 Å of a lithium ion count as its first shell; this is the cut-off the platform uses for lithium-anion contacts and the first minimum of the lithium-polymer pair distribution. As noted above, these atoms are of the types the platform lists as solvating but are not always the ones that coordinate lithium, so the shell describes the local polymer environment of each ion rather than its coordination. The persistence of the shell is the intermittent correlation of these contacts at a given lag, and ion aggregation comes from the cluster populations distributed with each trajectory. Rank correlations are taken over all long runs (replica noise: the 92 replicated systems with a long run); bold values have a 95% bootstrap interval that excludes zero.

Table S7: Lithium solvation and aggregation in the long runs, and their rank correlation with the long-run diffusivity, the overestimation by the short run ($\ln D_5 - \ln D_{50}$), its rank error and the noise between replicas.

	Median		Rank correlation with			
Descriptor	Screening	Literature	$\ln D$, 50 ns	Overestimation	Rank error	Replica noise
Polymer atoms in the first shell	2.0	3.4	+0.08	+0.07	-0.06	-0.04
Li^+ without polymer contact	7%	1%	-0.09	-0.06	+0.06	+0.08
Polymer contacts kept after 1 ns	59%	74%	+0.11	+0.09	-0.01	-0.10
Polymer contacts kept after 5 ns	54%	66%	+0.01	+0.14	+0.04	-0.08
Free Li^+ (no anion)	31%	28%	-0.02	-0.17	-0.06	+0.08
Li^+ in clusters of ≥ 5 Li^+	2%	1%	**-0.25**	**+0.26**	**+0.18**	+0.18

S6 End-point estimator

The end-point value of the 5 ns runs deviates from the power law fitted over 20% to 60% of the same curve with a standard deviation of 0.19 in log units, a typical factor of 1.21 (Fig. S1); 5.4% of the systems deviate by more than a factor of 1.5, and the rank correlation between the end-point estimate and the origin-averaged estimate at half the run length is 0.92. This deviation combines the single-origin noise of the end point with any departure of the curve from a power law; the replicated runs (Table S6) give the direct comparison of the two estimators.

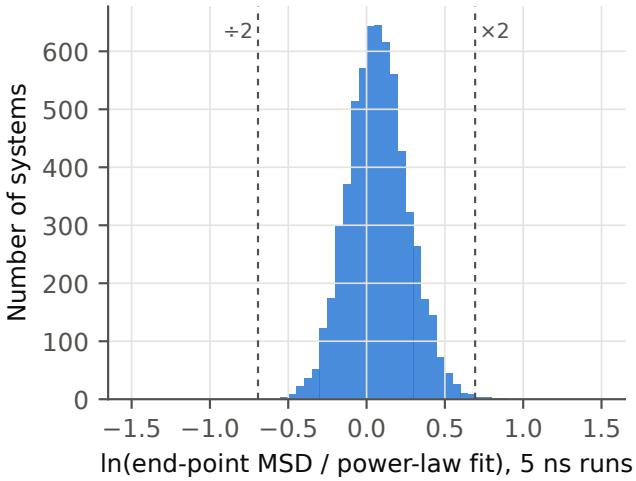


Figure S1: Deviation of the end-point MSD of the 5 ns runs from the power law fitted over lags between 20% and 60% of the run. Dashed lines mark a factor of two.

S7 Paired short and long runs

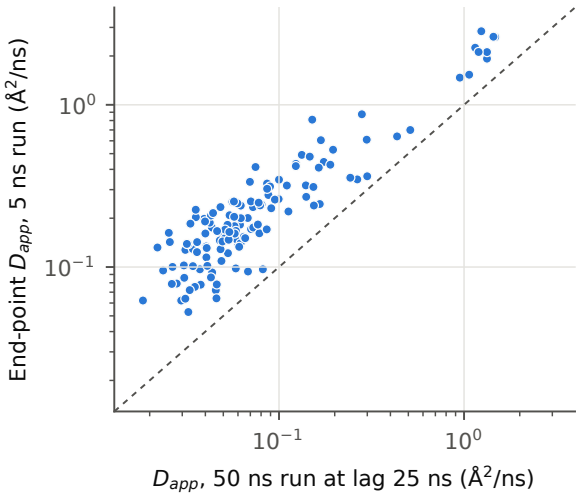


Figure S2: End-point apparent diffusivity from the 5 ns run against the origin-averaged apparent diffusivity at a lag of 25 ns from the 50 ns run of the same system, which starts from the same initial structure (133 systems). The dashed line is the identity.

S8 Prefix-based estimators and stopping rules

Table S8: Estimates from a prefix of length T relative to the reference $D_{\mathrm{app}}(25\,\mathrm{ns})$ of the full 50 ns run: median ratio and typical multiplicative error (133 runs).

	End-point		Half-lag		Linear fit		Power-law extrapolation	
T (ns)	Ratio	Error	Ratio	Error	Ratio	Error	Ratio	Error
1	6.56	×6.56	10.34	×10.34	3.22	×3.27	0.53	×1.87
2	4.43	×4.43	6.39	×6.39	2.83	×2.85	0.72	×1.50
5	2.83	×2.83	4.05	×4.05	2.02	×2.02	0.95	×1.37
10	2.02	×2.02	2.55	×2.55	1.48	×1.51	1.07	×1.26
20	1.43	×1.45	1.78	×1.78	1.16	×1.24	1.11	×1.20
30	1.20	×1.23	1.38	×1.38	0.95	×1.20	1.05	×1.13
50	0.89	×1.13	1.00	×1.00	0.73	×1.37	0.97	×1.03

The regime rules rarely fire: the effective exponent of a prefix reaches 0.9 before 50 ns in 3% of the runs and 0.8 in 19%, so they save almost no simulation, and when the $\beta \geq 0.8$ rule fires the error of the runs that stop is 1.39. The precision rule with a 20% tolerance stops early in 80% of the runs but with a typical error of 2.22 for those runs: a small bootstrap CV measures the precision of a lag-dependent quantity, not its distance from the long-lag value.

Table S9: Sensitivity of the prefix-based comparison to the reference (133 long runs). Each cell gives the typical multiplicative error and, in parentheses, the median ratio of the estimate to the reference. The power law is extrapolated to the quantity that defines each reference: the apparent diffusivity at that lag, or the mean slope of $A\tau^{\beta}$ over 20 to 40 ns.

Reference (from the 50 ns run)	End-point, 5 ns	Linear fit, 5 ns	Power law, 5 ns	Power law, 10 ns
D_{app} at 15 ns	×2.37 (2.37)	×1.80 (1.73)	×1.33 (1.06)	×1.22 (1.16)
D_{app} at 25 ns (main)	×2.83 (2.83)	×2.02 (2.02)	×1.37 (0.95)	×1.26 (1.07)
D_{app} at 35 ns	×3.15 (3.15)	×2.23 (2.23)	×1.39 (0.86)	×1.32 (1.00)
D_{app} at 40 ns	×3.27 (3.27)	×2.33 (2.33)	×1.45 (0.83)	×1.30 (0.95)
Slope of MSD, 20 to 40 ns	×4.01 (4.01)	×3.05 (3.05)	×1.99 (0.54)	×1.52 (0.68)

Table S10: Stopping rules evaluated on the long runs (133 runs). Runs that do not meet a rule before 50 ns are simulated for the full length and evaluated there. Mean time and the first error column are averaged over all runs; the last column restricts the error to the runs that stop early.

Rule	Estimator	Stop before 50 ns	Mean time (ns)	Error, all runs	Error, runs that stop
Regime: $\beta \geq 0.9$	Linear fit	3%	49.3	×1.37	×1.38
Regime: $\beta \geq 0.8$	Linear fit	19%	44.6	×1.39	×1.39
Precision: CV < 10%	Linear fit	2%	49.3	×1.37	×1.36
Precision: CV < 20%	Linear fit	80%	15.6	×1.79	×2.22
Precision: CV < 10%	Power law	24%	43.4	×1.04	×1.13
Fixed 5 ns	End-point	n/a	5.0	×2.83	n/a
Fixed 5 ns	Linear fit	n/a	5.0	×2.02	n/a
Fixed 10 ns	Linear fit	n/a	10.0	×1.51	n/a
Fixed 20 ns	Linear fit	n/a	20.0	×1.24	n/a
Fixed 5 ns	Power law	n/a	5.0	×1.37	n/a
Fixed 10 ns	Power law	n/a	10.0	×1.26	n/a
Fixed 20 ns	Power law	n/a	20.0	×1.20	n/a

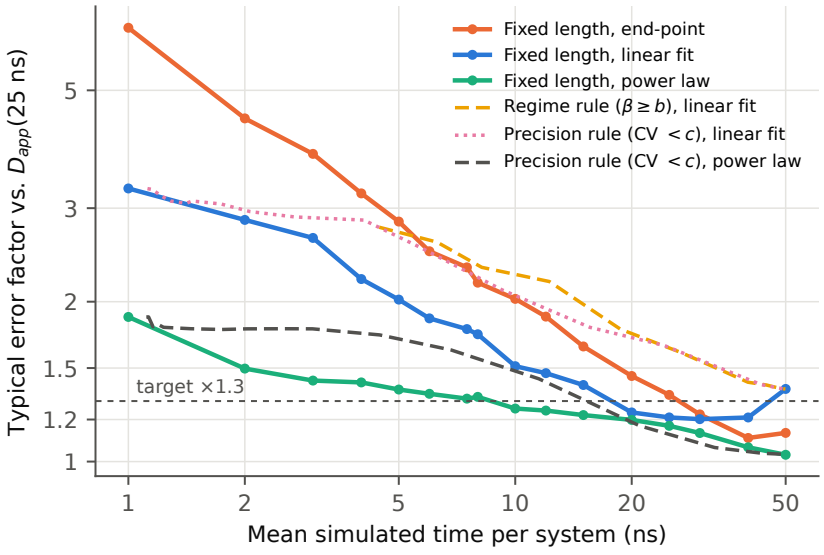


Figure S3: Cost-error frontier of estimators and stopping rules on the 133 long runs. Each line follows one rule as its parameter varies; runs that do not stop before 50 ns count as 50 ns. Fixed-length power-law extrapolation forms the lower envelope: a typical error of ×1.5 requires 2.0 ns with it, against 12.0 ns with a linear fit and 20.0 ns with the end-point estimate (10.0, 20.0 and 30.0 ns for ×1.3).

Table S11: Settings chosen on half of the systems and evaluated on the other half (median over 500 random partitions): mean simulated time, typical error, and percentage of partitions in which the held-out half meets the target; n/a means that no setting met the target on the half used for the choice.

	Target error ×1.5			Target error ×1.3		
Rule	Time (ns)	Error	Target met	Time (ns)	Error	Target met
Fixed length, end-point	20.0	×1.45	97%	30.0	×1.23	84%
Fixed length, linear fit	12.0	×1.47	61%	20.0	×1.24	96%
Fixed length, power law	2.0	×1.50	59%	7.5	×1.31	45%
Regime rule, linear fit	35.2	×1.48	64%	n/a	n/a	0%
Precision rule, linear fit	37.0	×1.47	67%	50.0	×1.40	0%
Precision rule, power law	11.9	×1.43	76%	20.9	×1.18	93%

S9 Adaptive allocation: algorithm and robustness

Algorithm. All quantities are in $\ln D$ of the end-point label. A campaign has $N = 6{,}137$ candidates, a target of the top k and a mean budget of b runs per candidate, each run starting from a new, equilibrated structure.

1. *Model.* System values $\mu_i \sim \mathcal{N}(0, \sigma_{\mathrm{sys}}^2)$ with $\sigma_{\mathrm{sys}} = 0.33$; a run of system i is $\mu_i + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \sigma(\mu_i)^2)$ and $\sigma(\mu) = \max\{a + \gamma\,\mu/\sigma_{\mathrm{sys}},\, 0.08\}$, $a = 0.35$, $\gamma = -0.067$, fitted so that the mean noise of the slower and faster halves matches the replicated systems.
2. *Posterior.* After n_i runs with mean $\bar{y}_i$, the posterior of μ_i is approximated by a normal with precision $1/\sigma_{\mathrm{sys}}^2 + n_i/\sigma(\bar{y}_i)^2$ and mean $(n_i\bar{y}_i/\sigma(\bar{y}_i)^2)$ divided by that precision. Because the noise depends on μ_i, the exact posterior is not normal; the approximation evaluates the noise at the observed mean (plug-in). Allocating instead with a constant-noise model, whose posterior is exactly normal, gives practically the same recall while the runs keep their mobility-dependent noise (see below), so the approximation does not drive the results.
3. *First round.* One run for every candidate.
4. *Batches.* While budget remains: take the cut-off c_k as the k-th largest posterior mean, compute $p_i = \Phi[(m_i - c_k)/s_i]$ from the posterior mean m_i and standard deviation s_i, and give one more run to the 5% of candidates with the largest $p_i(1 - p_i)$.
5. *Selection.* The top k by posterior mean. The metric is the fraction of the true top k, the k largest μ_i, that is selected.

The true top set is therefore defined by the expected 5 ns label of each system, not by its long-time diffusivity. Uniform replicates, successive halving and the two-stage re-screening use the same model, posterior and selection; only the choice of who receives each run differs. All random draws use fixed seeds, given in the code.

Table S12: Recall of the true top set for each allocation and mean budget (end-point label, Gaussian system values, 200 simulated campaigns of 6,137 candidates); the values plotted in the allocation figure of the main text. Each run starts from a new, equilibrated structure. With one run per candidate the recall of the top 10% is 48%, not the 45% of the noise-ceiling table, because here the noise depends on mobility.

		Mean runs per candidate					
Target	Allocation	1	1.5	2	3	4	6
Top 1%	Uniform replicates	32%	n/a	52%	63%	67%	73%
	Successive halving	32%	81%	85%	88%	90%	91%
	Boundary-focused	32%	75%	82%	87%	89%	92%
Top 5%	Uniform replicates	41%	n/a	57%	65%	70%	75%
	Successive halving	41%	70%	77%	82%	85%	89%
	Boundary-focused	41%	70%	77%	83%	86%	89%
Top 10%	Uniform replicates	48%	n/a	61%	68%	72%	77%
	Successive halving	48%	65%	73%	79%	83%	86%
	Boundary-focused	48%	68%	75%	81%	84%	87%

The saving for the top 10% against four uniform replicates is $\times 2.2$ when the system values follow the empirical shape of the 6,137 short runs instead of a normal distribution, and $\times 2.4$ with the origin-averaged label. Using the mobility-dependent noise in the allocation does not help: the recall of the top 10% with four runs per candidate is 84% with it and 85% with a constant noise. Successive halving with restarted rounds performs similarly to boundary-focused

Table S13: Mean number of 5 ns runs per candidate that boundary-focused allocation needs to match the recall of uniform replicates, and the corresponding saving (end-point label, Gaussian system values, 200 simulated campaigns of 6,137 candidates).

	Uniform replicates matched		
Target	2 runs	3 runs	4 runs
Top 1%	1.1 (×1.7)	1.2 (×2.5)	1.2 (×3.2)
Top 5%	1.2 (×1.7)	1.3 (×2.2)	1.5 (×2.7)
Top 10%	1.2 (×1.6)	1.5 (×2.0)	1.8 (×2.2)

allocation. With the end-point label and 1.25 runs per candidate it recovers 74% of the top 1%, against 68%; with the origin-averaged label and four runs per candidate it recovers 84%, against 89%.

Table S14: Recall of the true top set with two and four runs per candidate on average, and saving against four uniform replicates, for boundary-focused allocation planned with the measured reliability or with the ends of its 95% interval (the runs always follow the measured noise), and for a two-stage re-screening with the best of three fixed shortlists ($2k$, $4k$, $8k$) chosen in hindsight (with the shortlist fixed at $4k$ in advance the savings are ×3.0, ×2.4 and ×1.8). End-point label, Gaussian system values, 200 simulated campaigns; Monte Carlo standard error of each recall at most 0.4 percentage points (median 0.1).

	Boundary-focused, planned with r =			Two-stage
Target	0.47 (true)	0.31	0.62	re-screening
Top 1%	82% / 89% (×3.2)	78% / 85% (×3.0)	83% / 90% (×3.2)	80% / 85% (×3.2)
Top 5%	77% / 86% (×2.7)	76% / 85% (×2.6)	76% / 86% (×2.6)	74% / 80% (×2.4)
Top 10%	75% / 84% (×2.2)	74% / 83% (×2.2)	74% / 83% (×2.1)	71% / 80% (×1.8)

Underestimating the reliability makes the posterior trust each run less and spreads the replicates more widely; overestimating it concentrates them. Neither changes the saving much (Table S14). If instead the true reliability, not only the planned one, lies at either end of its interval (keeping the total spread of the short runs, and with the allocation using the correct value), the saving against four uniform replicates is ×2.4 ($r = 0.31$) and ×3.2 ($r = 0.62$) for the top 1%, ×2.1 and ×2.9 for the top 5%, and ×1.8 and ×2.5 for the top 10%: noisier labels reduce the saving but do not remove it. The two-stage re-screening is as good as boundary-focused allocation for the top 1%, where the best shortlist is $8k$, but falls behind for broader targets, and a shortlist that is too short cannot recover candidates that the first run ranked low.

S10 Check of the adaptive allocation on real runs

The long runs start from the same structure as the first runs, so they favour the first run and understate the value of any second run from a new structure; this is why no strategy improves on the first run alone for the top 20%. The size of this bias is not known, so the check is probably, but not demonstrably, conservative.

Table S15: Check of the adaptive allocation on the replicated systems. Calibration: 94 systems, predictions in both directions (first run → second and second → first), noise parameters estimated leaving out the system evaluated. Campaign: 92 systems with a long run; every system receives its first run and 46 receive a second run, chosen at the boundary or at random (mean of 2000 draws); recall of the top set according to the long run. Interval: bootstrap over systems of the difference between boundary and random placement.

	Top 10%	Top 20%
Brier improvement over the base rate	36%	16%
Calibration slope (95% CI)	1.10 (0.66 to 2.27)	0.63 (0.33 to 1.36)
Systems in the top set, k	9	18
Recall, first runs only	67%	72%
Recall, second runs at random	71%	70%
Recall, second runs at the boundary	78%	72%
Recall, second runs for all	78%	72%
Boundary − random (95% CI, points)	−11 to +22	−17 to +11